

UTM

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FACULTY OF COMPUTING
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Project: Phase 2
Database Conceptual Design (ERD)

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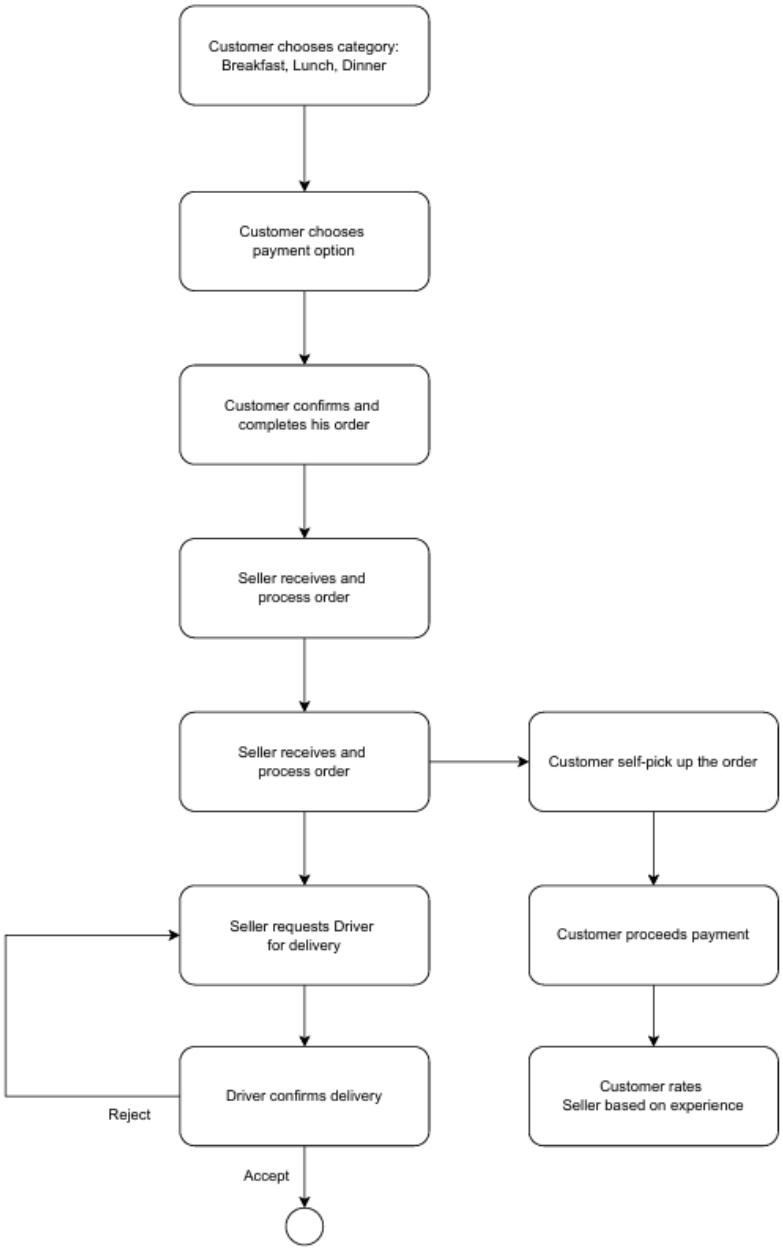
1.0 Introduction

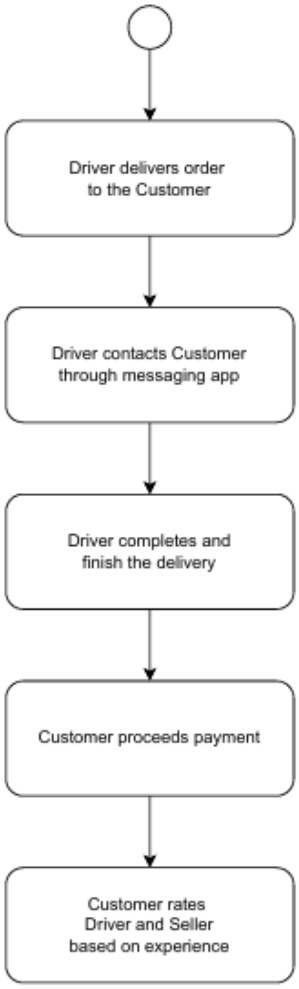
Our project, UTMart, is an online system developed for students, sellers and drivers in Universiti Teknologi Malaysia, Johor Bahru. This system is built to replace the conventional way of buying, selling and promoting foods inside the campus. Sellers would usually promote and sell their foods in common messaging apps such as Whatsapps or Telegram. But with this system, they can promote and sell their foods conveniently and efficiently by the tip of their finger. This system will utilize a database system where we can oversee the data that is stored inside the database system.

However, in order to develop this system, it requires the database system to function well for a more efficient and convenient process. Before developing the actual database, we have to go through some steps where we plan the database, define the system, collect and analyse requirements then lastly, we design the database. There are three phases in database design which are, conceptual database design, logical database design and physical database design.

In this report, we are focusing on the conceptual database design where we construct the entity relationship diagram (ERD) and data dictionary table. Entity relationship diagram (ERD) is basically a flowchart that shows how every entity relates with each other in the database while data dictionary is a description table for entities involved in the database. This step is very important in database design because it will help in visualizing the relations and attributes needed in designing the database.

2.0 Flowchart (Proposed System)




Driver delivers order
to the Customer
Driver contacts Customer
through messaging app
Driver completes and
finish the delivery
Customer proceeds payment
Customer rates
Driver and Seller
based on experience

3.0 Data & Transactional Requirement

3.1 Business Rule

In UTMart system business rule:

1. The restaurants in each arked start working at 7.00AM to 10.00PM (15 hours).
2. Each customer is required to log in or register for an account to place an order.
3. Many customers can place orders once the restaurant is opened.
4. Customers can place at least one order and the order has limitations (depends on the food availability).
5. Customers can make requests to their order.
6. Customers are forbidden from ordering items that are unavailable.
7. Customers can only choose one payment method, cash-on-delivery (COD) or QR payment.
8. Each order has order information, and each payment is made for one order.
9. Each order can be delivered by one driver, each driver can deliver many orders.
10. Customers can choose for delivery or self-pickup.
11. If no delivery driver is available after 30 minutes, the customer can choose to pick-up the order themself.
12. Each delivery driver can deliver up to 5 location for each maximum order with details:
    Delivery within 1 km = RM1,
    Delivery within 2 km = RM2,
    Delivery from 3 km and further = RM3
13. Each customer can leave feedback and the feedback can be seen by many customers.
14. Each sold food and transaction made must be recorded

3.2 Data & Transaction

3.2.1 Data Entry

- Input customer details such as name, sex, date of birth, email, matric/passport number and contact information
- Input seller details such as name, position and banking information (bank account and DuitNow QR code).
- Input food details like menu items.

3.2.2 Data Update/Delete

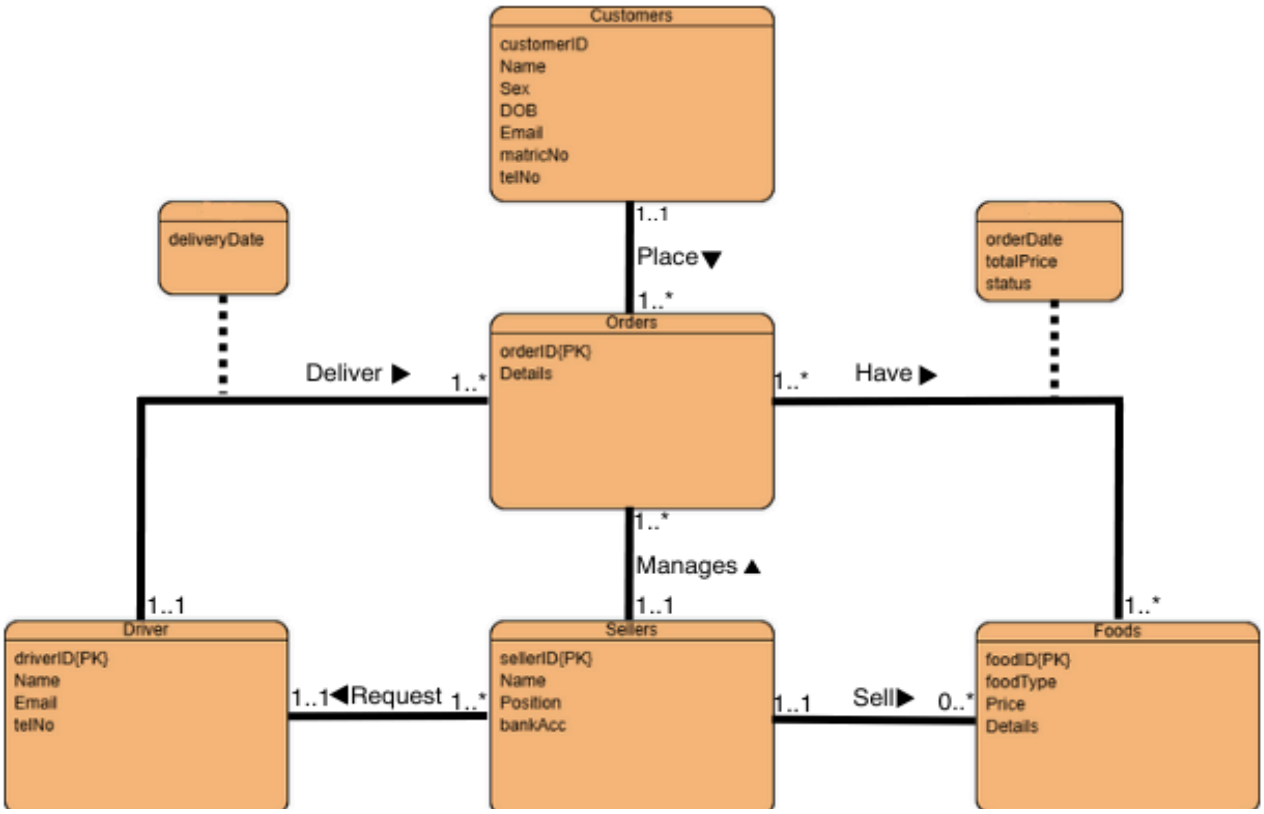
- Update/Delete menu items
- Update food price and availability
- Update order status
- Update delivery address
- Update/delete the details of the payment

3.2.3 Data Queries

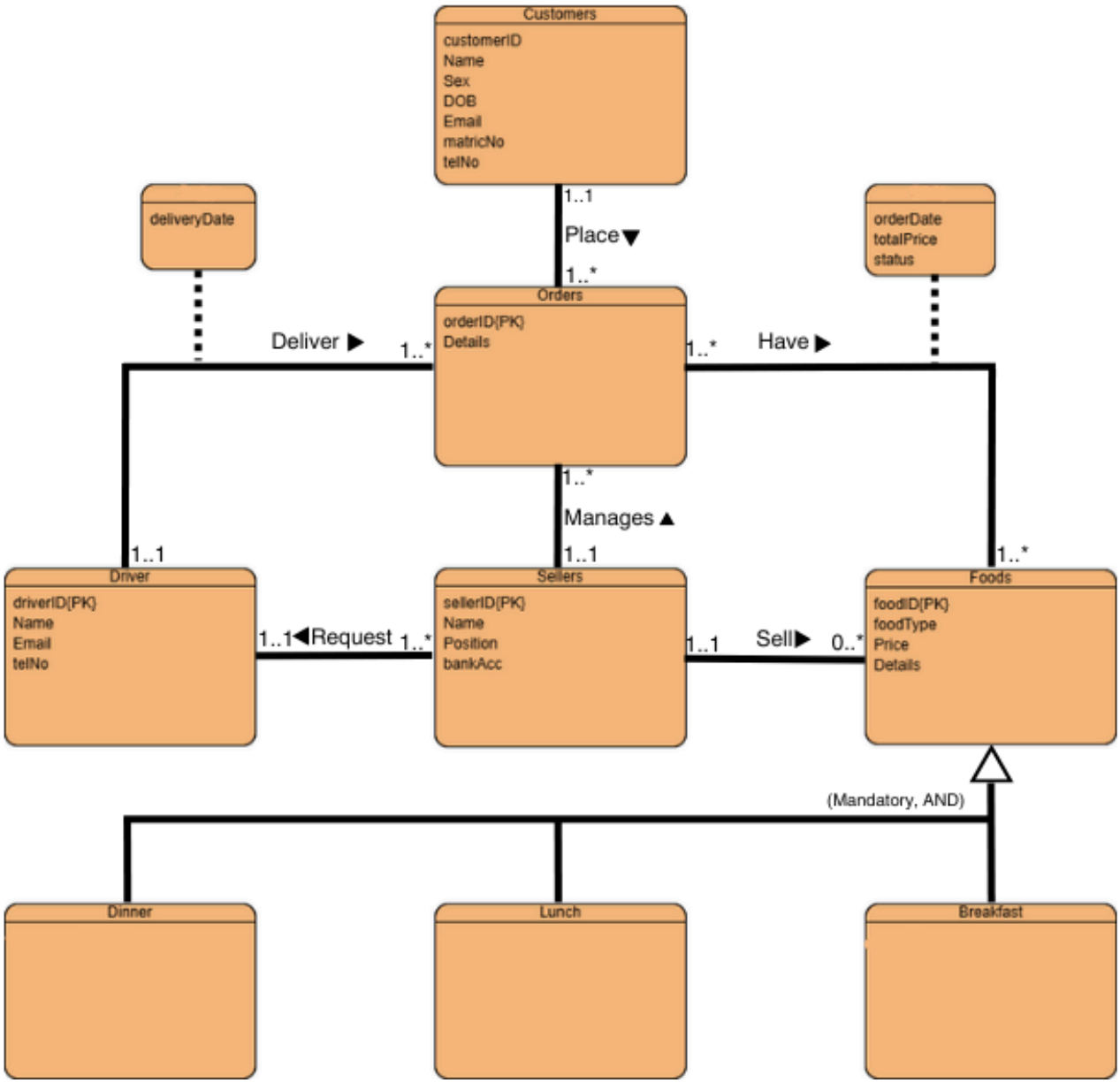
- List the restaurant details
- List details of food
- Identify order status
- List order status
- List active promotions or discounts offered by restaurants.
- List delivery driver details
- Identify the payment status

4.0 Database Conceptual Design (ERD)

4.1 Conceptual ERD



4.2 Enhanced ERD (EERD)



5.0 Data Dictionary

5.1 Data Dictionary for Entities

Entity Name	Description	Aliases	Occurrence
Customers	Term for all users who are not selling or promoting any food in the system.	Buyer	Each customer can place zero or more orders.
Orders	Term for describing all orders from customers.	Request	Each order can be placed by only one customer. Each order can be delivered by only one driver. Each order can be received by only one seller. Each order can have one or many food
Foods	General term for describing a meal that is being sold by sellers.	Meal	Each food can be sold by only one seller.
Sellers	General term for describing all seller details.	Merchant	Each seller can receive one or more orders. Each seller can request one or more drivers for delivery. Each seller can sell one or more foods.
Drivers	Term for all drivers who deliver food to customers.	Rider	Each driver can deliver one or more orders.

5.2 Data Dictionary for Relationship

Entity Name	Multiplicity	Relationship	Entity Name	Multiplicity	Attributes
Customers	1..1	Place	Orders	0..*	
Orders	1..*	Have	Foods	1..*	orderDate totalPrice status
Sellers	1..1	Manage	Orders	1..*	
Sellers	1..*	Request	Drivers	1..*	
Sellers	1..1	Sell	Foods	0..*	
Drivers	1..1	Deliver	Orders	1..*	deliveryDate

5.3 Data Dictionary for Attributes

Entity Name	Attributes	Description	Data Type & Length	Nulls	Multi-valued
Customers	customerID	Uniquely identifies a customer	5 variable characters	No	No
	Name	Name of the customer	30 variable characters	No	No
	Sex	Gender of customer	1 character (M or F)	No	No
	DOB	Date of birth of customer	Date	Yes	No
	Email	Email of the customer	15 variable characters	No	No
	matricNo	Unique number to identifies student	10 variable characters	No	No
	telNo	Customer's phone number	10 integers	No	Yes
Orders	orderID	Uniquely identifies an order	5 variable characters	No	No
	Details	Describing the order details	50 variable characters	Yes	No
Foods	foodID	Uniquely identifies a food	5 variable characters	No	No
	foodType	Determine food categories	1 character (I, F or R)	No	No
	Price	Price of the food	Decimal(4,2)	No	No
	Details	Additional description about the food	100 variable characters	Yes	No
Sellers	sellerID	Uniquely identifies a seller	5 variable character	No	No
	Name	Name of a seller	30 variable characters	No	No

	Position	Identifies the role of seller in UTM	10 variable characters	No	No
	bankAcc	Bank account number of seller	15 integers	No	Yes
Drivers	driverID	Uniquely identifies a driver	5 variable characters	No	No
	Name	Name of a driver	30 variable characters	No	No
	Email	Email of a driver	15 variable characters	No	No
	telNo	Driver's phone number	10 integers	No	Yes

6.0 Summary

In conclusion, the system provides a clear view of the processes involved in data handling and data transaction. It also provides a detailed overview of the key business rules of UTMart. The flowchart shows how the customers are going to use UTMart in their daily life. UTMart operates under a strict business hour where the system will operate together with the working hours of restaurants in UTM. UTMart allows users to login and register an account. The system also allows the user to place an order, allow the seller to request for a driver and allow the driver to deliver the order. The system will request for necessary data for the customer, seller and food details. The system also allows the database of the system to be updated or deleted. The identified entities in the system are customers, orders, foods, sellers, and drivers and each entity is related to one another with their respective multiplicity to ensure the clarity of relationship between the entities. For instance, customers can place orders, drivers will deliver orders, sellers will receive orders, sellers sell food and orders have foods.